

Spatial Health: How Smoking Shapes Our Neighborhoods?

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INTRODUCTION

- Neighborhoods are segregated not only by income, but also by health.
 - Within MSAs, LE is about 7 years higher in top- vs. bottom-decile income tracts.
- Patterns of income segregation are well understood
Benabou (1996), Durlauf (1996), Fernandez and Rogerson (1996, 1998)
- But less is known about the drivers of health segregation.
- Understanding what drives spatial health inequality matters:
 - Health is a major component of welfare
 - Health and income segregation may reinforce each other, creating local opportunity traps.

WHAT COULD EXPLAIN THESE SPATIAL HEALTH GAPS?

- We focus on smoking: it starts early, persists, and strongly predicts longevity.
- Study two mechanisms linking neighborhoods to children's health:
 - **Parental sorting**
 - Smoking parents sort into specific neighborhoods creates health segregation
 - Intergenerational persistence in smoking transmits these health gaps across generations.

Göhlmann et al. (2010)

- **Local externalities**
 - Children are exposed to local smoking behavior.
 - Exposure can shape their own smoking decisions.

Norton (1998); Powell et al. (2005); Arduini et al. (2025)

THIS PAPER

- We develop an OLG model with multiple neighborhoods.
- Teens and parents make three decisions that shape health:
 - 1 Parents decide where to live
 - 2 Kids decide whether to attend college
 - 3 Kids decide whether to smoke
- Local externalities:
 - Health Behaviors: ↑ local smoking rate increases the probability of smoking.
 - Education: ↑ local college rate (better schools/role models) raises college attendance.
- House prices adjust to clear neighborhood markets.

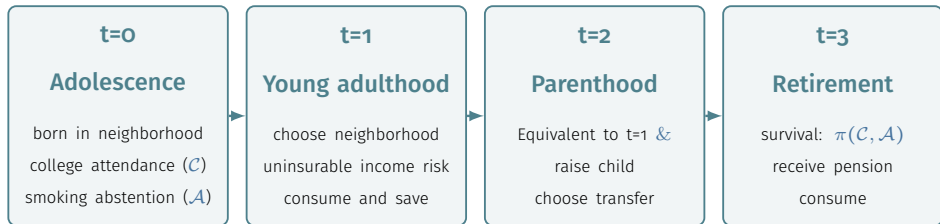
THIS PAPER

- We estimate the model using data on the US: AddHealth, PSID, and HRS.
- We discipline spillover parameters using reduced-form exposure effects.
- We use the model to decompose neighborhood LE gaps:
 - How much comes from parental sorting?
 - How much comes from local externalities in education and smoking?
- We quantify feedback between health and education:
 - How much health segregation is mediated by education externalities?
 - How much income segregation is mediated by smoking externalities?
- We evaluate place-based policies through welfare and longevity gains.

LITERATURE

- Understanding the cost of bad health and health inequalities across SES
Cole et al. (2019); Mahler and Yum (2023); Margaritis and Wallenius (2023); Hosseini et al. (2025)
→ We focus on the spatial dimension to understand health inequalities
- Recent quantitative literature on the welfare consequences of income segregation
Chyn and Daruich (2025); Fogli et al. (2025); Eckert and Kleineberg (2025)
→ We analyze how health drives economic segregation and welfare
- Health in space
Margaritis and Wallenius (2025)
→ Equilibrium model where spillovers are endogenous

MODEL



one model period = 20 years

PREFERENCES

- Individuals discount the future at a rate β
- The utility function is given by:

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma} + b,$$

where b is a constant capturing the joy of living.

Hall and Jones (2007)

- Richer individuals value longevity more because they have higher consumption flows.

ADOLESCENCE

- A teenager is born in a given neighborhood n .
- Receives a parental transfer a_0 .
- Chooses college attendance and whether to smoke.
- Chooses current consumption and next-period assets a' .
- Teens are heterogeneous in the utility of abstaining from smoking and attending college.
- Utility depends on:
 - Parental background: college (C_p) and abstinence (A_p)
 - Neighborhood characteristics: share of non-smokers ($\bar{A}_n$) and share of college graduates ($\bar{C}_n$)
 - Idiosyncratic shocks to college (ϵ_C) and abstinence (ϵ_A)

ADOLESCENCE

- The teenager solves:

$$\begin{aligned}
 V_0(a_0, \mathcal{C}_p, \mathcal{A}_p, \epsilon_{\mathcal{C}}, \epsilon_{\mathcal{A}}, n) = & \max_{\substack{c, a' \\ \mathcal{C}, \mathcal{A} \in \{0,1\}}} \left\{ u(c) + \mathcal{C} u^{\mathcal{C}}(\mathcal{C}_p, \bar{\mathcal{C}}_n, \epsilon_{\mathcal{C}}) \right. \\
 & \left. + \mathcal{A} u^{\mathcal{A}}(\mathcal{A}_p, \bar{\mathcal{A}}_n, \epsilon_{\mathcal{A}}) + \beta \mathbb{E} V_1(a', \mathcal{C}, \mathcal{A}, z', n) \right\} \\
 \text{s.t.} \quad & c + a' = a_0 - \mathcal{C} \kappa_{\mathcal{C}} + (1 - \mathcal{C}) y_0 \\
 & \underline{a} \leq a' \leq 0
 \end{aligned}$$

ADOLESCENCE

- The utility shifters are given by:

$$u^C(\mathcal{C}_p, \bar{\mathcal{C}}_n, \epsilon_C) = \tau_0^C + \tau_p^C \mathcal{C}_p + (\tau_n^C + \tau_{pn}^C \mathcal{C}_p) \bar{\mathcal{C}}_n^{\rho_C} + \epsilon_C$$

$$u^A(\mathcal{A}_p, \bar{\mathcal{A}}_n, \epsilon_A) = \tau_0^A + \tau_p^A \mathcal{A}_p + (\tau_n^A + \tau_{pn}^A \mathcal{A}_p) \bar{\mathcal{A}}_n^{\rho_A} + \epsilon_A$$

$$\epsilon_C \sim \mathcal{U}[-\sigma^C, \sigma^C], \quad \epsilon_A \sim \mathcal{U}[-\sigma^A, \sigma^A],$$

with ϵ_C and ϵ_A independent.

YOUNG ADULthood AND PARENTHOOD

- Working life is divided into two periods:
 - 1 **Young adulthood**: agents first choose a neighborhood, then enter the labor market.
 - income depends on education, smoking status, and a persistent shock (z)
 - agents consume, save, and pay rent, cigarette costs, and medical expenditure
 - 2 **Parenthood**: + raise a child, and choose a wealth transfer.

YOUNG ADULTHOOD AND PARENTHOOD

Neighborhood Choice

- At the beginning of period 1 and period 2, the value function is:

$$V_t(a, \mathcal{C}, \mathcal{A}, z, n) = \mathbb{E} \left[\max_{n'} \left\{ \tilde{V}_t^{n'}(a, \mathcal{C}, \mathcal{A}, z, n) + \xi_{n'} \right\} \right],$$

where $\tilde{V}_t^{n'}(a, \mathcal{C}, \mathcal{A}, z, n)$ is the neighborhood-specific value function.

- We assume $\xi_{n'}$:
 - i.i.d. across neighborhoods and periods
 - type-I extreme value distributed with scale parameter σ_n .

PARENTHOOD

- The neighborhood-specific value function during parenthood is given by:

$$\tilde{V}_2^{n'}(a, \mathcal{C}, \mathcal{A}, z, n) = \max_{c, a', a_0^k} u(c) + \mu_{n'} - \mathbf{1}_{\{n' \neq n\}} \cdot \eta_2$$

$$+ \beta \pi(\mathcal{C}, \mathcal{A}) \mathbb{E}_{z'} \left[V_3(a', \mathcal{C}, \mathcal{A}, z', n') \right]$$

$$+ \alpha \mathbb{E}_{\epsilon_{\mathcal{C}}, \epsilon_{\mathcal{A}}} \left[V_0(a_0^k, \mathcal{C}, \mathcal{A}, \epsilon_{\mathcal{C}}, \epsilon_{\mathcal{A}}, n') \right]$$

$$\text{s.t. } c + a' + a_0^k = T(y_2(\mathcal{C}, \mathcal{A}, z)) + (1 + r)a - 2P_{n'} - m(\mathcal{C}, \mathcal{A}) - (1 - \mathcal{A})\tau_{\text{cig}},$$

where

η_2 : non-pecuniary moving cost

$P_{n'}$: rental price in neighborhood n'

$m(\mathcal{C}, \mathcal{A})$: medical expenditure

$\mu_{n'}$: amenity value

α : altruism parameter

a_0^k : initial assets of the kid

$\pi(\mathcal{C}, \mathcal{A})$: survival probability

τ_{cig} : cigarette cost

RETIREMENT

- Retirees receive pensions, pay housing and medical bills, and consume savings.
- Pensions depend on education, smoking, and last productivity shock.

$$V_3(a, \mathcal{C}, \mathcal{A}, z, n) = u(c) + \mu_n$$

$$\text{s.t. } c = T(\omega(\mathcal{C}, \mathcal{A}, z)) + (1 + r)a - P_n - m(\mathcal{C}, \mathcal{A}) - (1 - \mathcal{A})\tau_{\text{cig}}$$

HOUSING MARKET

- Housing prices are determined in equilibrium.
- The supply of housing is given by:

$$S_n = \Psi_n P_n^{\tau_h},$$

where τ_h drives the price elasticity of housing supply, Ψ_n are housing supply shifters

EQUILIBRIUM

- A stationary equilibrium is a set of:
 - household policies for education, smoking, savings, consumption, transfers, and location;
 - house prices $\{P_n\}_{n \in \{A, B, C\}}$;
 - a stationary distribution of adults across states and neighborhoods;
 - local neighborhood environments $(\bar{C}_n, \bar{A}_n)$.
- such that households optimize taking prices and local environments as given, and:

$$D_n = \Psi_n P_n^{\tau_h} \quad \forall n,$$

$$\bar{C}_n = \frac{\int \mathbf{1}\{n_i = n\} \mathcal{C}_i d\lambda^p(i)}{\int \mathbf{1}\{n_i = n\} d\lambda^p(i)}, \quad \bar{A}_n = \frac{\int \mathbf{1}\{n_i = n\} \mathcal{A}_i d\lambda^p(i)}{\int \mathbf{1}\{n_i = n\} d\lambda^p(i)}.$$

- The local environments are therefore consistent with the college and non-smoking choices of parents living in each neighborhood.

DATA

- To estimate the model we are going to use AddHealth data.
- Tracks a representative cohort of US adolescents
 - $t = 0$: 1994-95 school year, grades 7-12.
 - $t = 1$: 2016-18, aged 34-40.
- Includes comprehensive data on the parent's smoking and college decisions.
- Contains tract-level statistics on neighborhood characteristics.
- Sample: 6,034 parent-child pairs.

DATA

- We use three neighborhoods: $n \in \{A, B, C\}$.
- Add Health kids are assigned by tract income rank within their MSA:
 - A : bottom 30%
 - B : 30%–90%
 - C : top 10%

ESTIMATION

- We follow a standard two-step procedure:

External:

- Wages $y_t(\mathcal{C}, \mathcal{A}, z)$ (PSID)
- Survival probabilities $\pi(\mathcal{C}, \mathcal{S})$ (HRS)
- College costs $\kappa_{\mathcal{C}} = 60k$
- House Prices (Add Health)
- Elasticity of housing supply: 1.75 (Saiz, 2010)
- Replacement rate: 0.494 (OECD, 2019)

ESTIMATION

Internal: The remaining parameters are estimated internally.

We use Add Health data to match:

- College graduation rates conditional on neighborhood and parental education
- Smoking rates conditional on neighborhood and parental smoking
- Parental transfers
- Value of Statistical Life
- Asset to income ratio
- Average marginal tax rate
- Intergenerational mobility

ESTIMATION

Estimated parameters: smoking and college decision

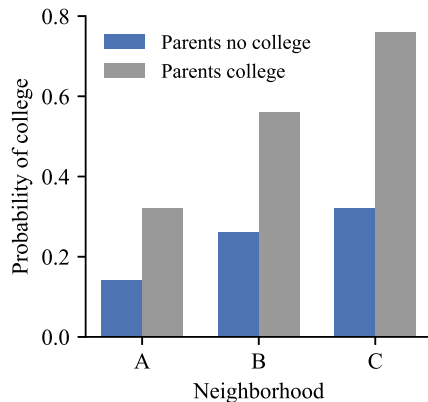
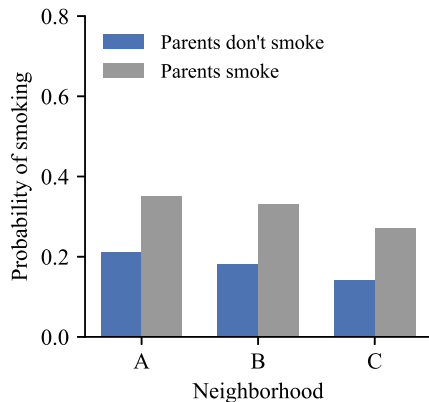
Parameter	Description	Value	
		$j = \mathcal{A}$	$j = \mathcal{C}$
τ_0^j	constant utility cost	-0.05	-0.55
τ_p^j	parental influence	0.29	0.90
τ_n^j	neighborhood influence	-0.68	-1.07
τ_{pn}^j	parent-neighborhood interaction	-0.00	-0.36
ρ_j	neighborhood-influence curvature	-0.98	-0.14
σ^j	support of ϵ^j	0.92	1.10

ESTIMATION

Remaining estimated parameters

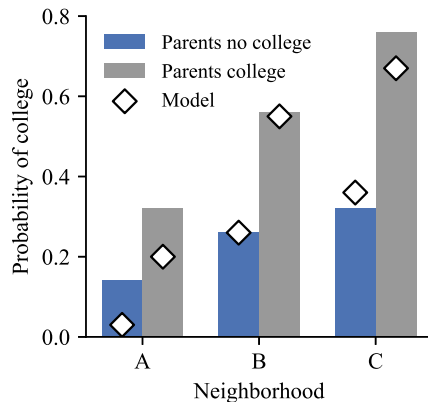
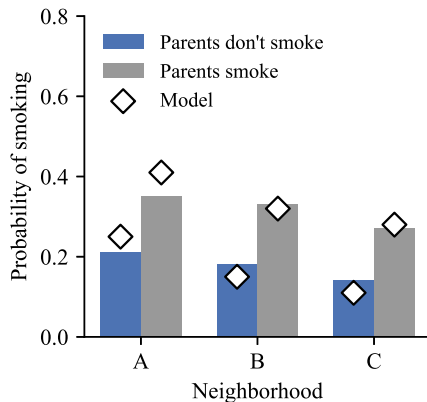
Parameter	Description	Value
b	value of life	12.30
β	annual discount factor	0.99
α	altruism parameter	0.51
μ_n	amenity in NBH B, C	0.07, 0.07
η_1, η_2	moving costs	0.03, 0.03
σ_n	Gumbel taste shock for NBH	0.04
λ	tax rate	0.77

ESTIMATION

Model Fit: Intergenerational Transmission of Smoking and College by Neighborhood

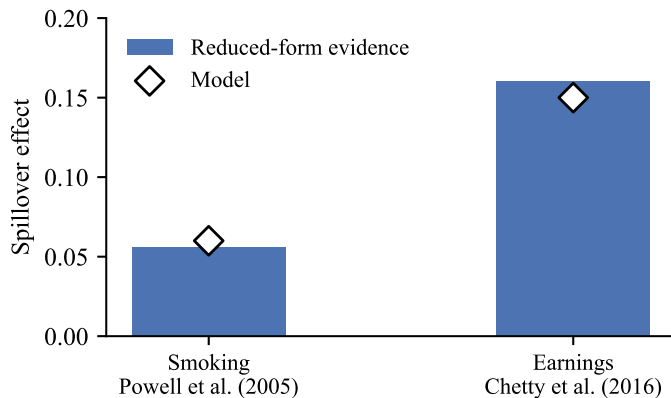
ESTIMATION

Model Fit: Intergenerational Transmission of Smoking and College by Neighborhood



ESTIMATION

Model Fit: Reduced-Form Exposure to Spillover Effects



ESTIMATION

Model Fit: Other Moments

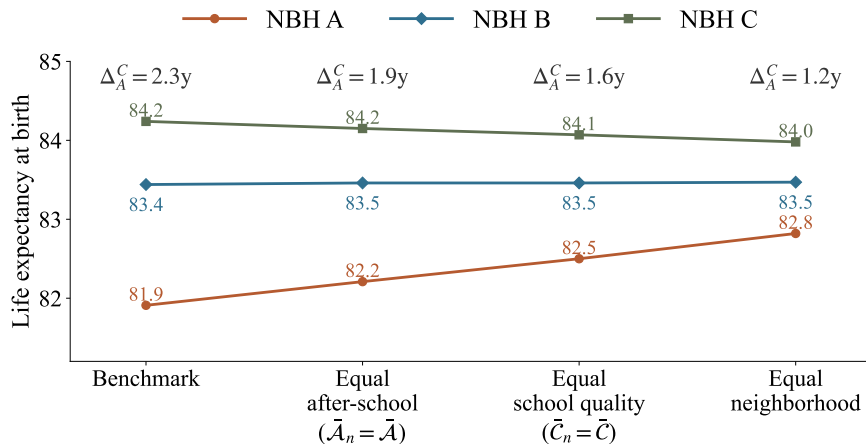
Moments	Model			Data		
VSL	\$5.9 million			\$6 millions		
A/Y at age 40	4.12			4.32		
Cost of raising a kid	170k			301k		
% of living in NBH A, B, C	0.24, 0.66, 0.10			0.30, 0.60, 0.10		
Average tax rate	0.34			0.35		
Prob. of staying in NBH	0.71			0.71		
Transition probabilities	$n_1 = A$	$n_1 = B$	$n_1 = C$	$n_1 = A$	$n_1 = B$	$n_1 = C$
$\Pr(n_1 \mid n_0 = A)$	0.54	0.41	0.05	0.49	0.48	0.03
$\Pr(n_1 \mid n_0 = B)$	0.26	0.69	0.05	0.25	0.69	0.06
$\Pr(n_1 \mid n_0 = C)$	0.29	0.53	0.18	0.18	0.67	0.16

PARENTAL SORTING VS. LOCAL EFFECTS

Decomposition

- We decompose neighborhood LE gaps at birth into:
 - **Local effects:** differences in smoking and education spillovers.
 - **Sorting:** differences in parental characteristics.
- For children born in each neighborhood, we recompute LE under:
 - **Equal school quality:** common education spillovers,
 - **Same after-school environment:** common smoking spillovers,
 - **Common local environment:** both spillovers equalized.
- The remaining gap captures parental sorting.

PARENTAL SORTING VS. LOCAL EFFECTS



Life Expectancy at Birth Across Neighborhoods: Decomposition

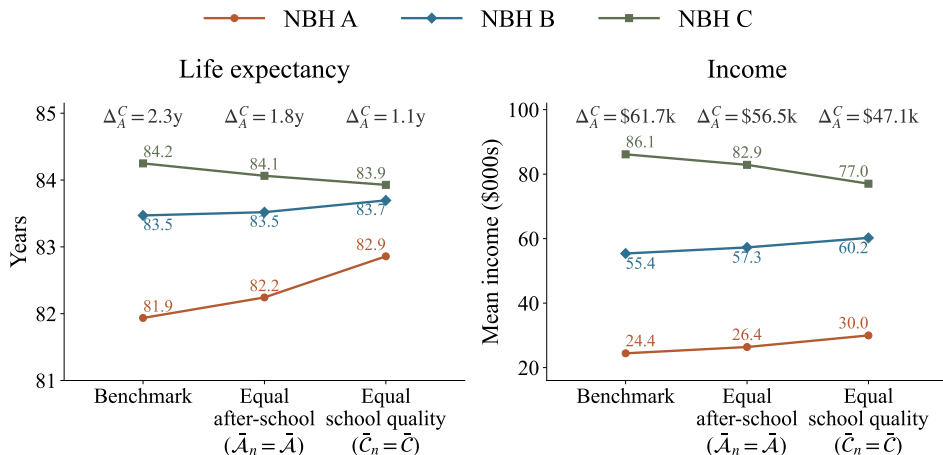
COUNTERFACTUAL EQUILIBRIA

When Sorting Responds

- The previous decomposition holds neighborhood choices fixed.
- In equilibrium, changing spillovers also changes where families choose to live:
 - Parents re-sort across neighborhoods.
 - House prices adjust to clear housing markets.
- We quantify how these responses reshape segregation:
 - How much school quality mediates health segregation?
 - How much smoking exposure mediates income segregation?

COUNTERFACTUAL EQUILIBRIA

Neighborhood Outcomes



WELFARE BENEFITS OF PLACE-BASED POLICIES

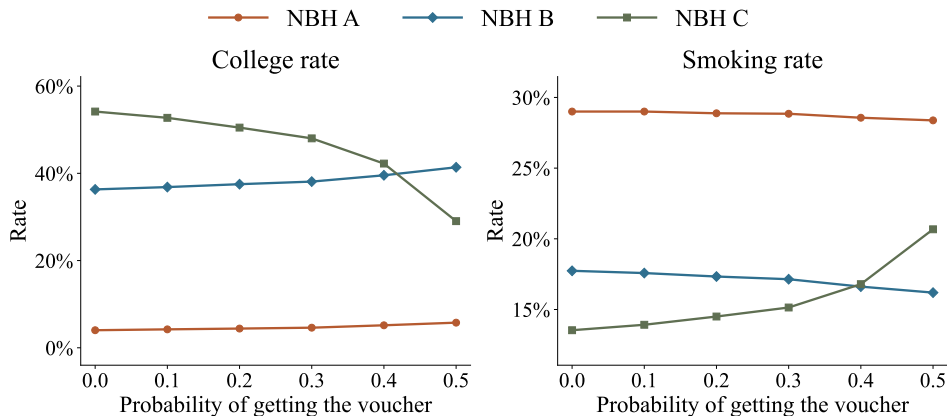
- We finally turn to evaluate the welfare gains from place-based policies.
- Decompose welfare gains of newborns into changes in:
 - consumption keeping longevity constant: $u(c_0) + \beta u(c_1) + \beta^2 u(c_2)$
 - longevity: $\beta^3 \pi(\mathcal{C}, \mathcal{A}) u(c_3)$

WELFARE BENEFITS OF PLACE-BASED POLICIES

Modeling

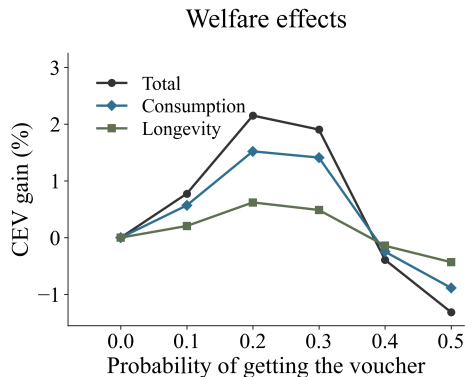
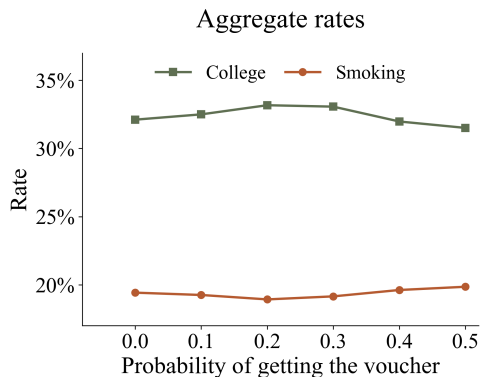
- At the beginning of period 2, some individuals become eligible for a housing voucher.
- Allows families to move to C at a subsidized price (MTO).
- Eligibility criteria:
 - Individuals in the bottom 20% of the income distribution.
 - Individuals living in neighborhood A .
- Scale of the policy:
 - Eligible individuals get a housing voucher with probability p
- We adjust the tax system to keep the government budget balanced.

WELFARE BENEFITS OF PLACE-BASED POLICIES



Neighborhood Characteristics in the Voucher Policy Counterfactuals

WELFARE BENEFITS OF PLACE-BASED POLICIES



Aggregate outcomes and Welfare Gains in the Voucher Policy Counterfactuals

CONCLUSIONS

- We develop a spatial equilibrium model with endogenous education, smoking and location decisions to understand the drivers of inequality in health across space.
- We find that 50% of the gap in life-expectancy across the top 10 and the bottom 30 percent in income distribution is explained by differences in parental characteristics.
- Local spill-overs explain the rest (mostly in education)
- 20% of the welfare gains from place-based policies come from changes in life-expectancy